

Roof Quantity Take-Off

Project
Sample (S2)

Roof System
60-mil TPO

Assembly
4 in Polyiso - R-24

Date
July 8, 2026

Field Area
2,511 SF
25.1 squares

Roof Perimeter
237 LF
Coping and base flashing run

Roof Assembly
TPO / 4 in ISO
R-24 - 2 layers - plywood deck

Roof Geometry

Front / bottom width	29 ft 5 in	29.42 ft
Left side	74 ft 6 in	74.50 ft
Right side	96 ft 3 in	96.25 ft
Sloped back edge	36 ft 7 in	36.58 ft
Top-right clip	5 ft 6 in	5.50 ft
Parapet height	4 ft 0 in	4.00 ft
Base flashing height	1 ft 6 in	18 in

*Modeled as right trapezoid: Field = $W \times (L_{left} + L_{right}) \div 2$.
Dimensions per sketch "Midland (S2)."*

Roof Assembly

Membrane	60-mil TPO - mechanically attached or fully adhered (see options)
Insulation	Two layers of 2 in polyiso (4 in total), joints staggered - R-24
Cover board	Optional 1/2 in HD-ISO / gypsum (recommended, not in totals)
Deck	Existing plywood
Parapets	4 ft high; TPO base flashing 18 in up wall; metal coping at top
Perimeter	Metal cap flashing (coping) over continuous termination bar

Quantity Take-Off

Item	Basis / Calculation	Qty	Unit	Notes
Membrane - 60-mil TPO				
TPO membrane - field	Field area x 1.10 waste	2,763	SF	Heat-welded seams
TPO base flashing - parapets	237 LF x 1.5 ft x 1.10	391	SF	18 in up wall; adhered
TPO rolls (10 ft x 100 ft)	(Field + flashing) / 1,000 SF	4	rolls	Field + base flashing
TPO cover strip	Detailing / seam allowance	1	roll	T-joints, patches
Insulation - Polyisocyanurate				
Polyiso - 2 layers (4 in total)	Field x 2 layers x 1.07	5,374	SF	R-24 LTTR; stagger joints
Polyiso boards (4 ft x 8 ft)	5,374 SF / 32 SF per board	168	boards	84 boards per layer
Insulation fasteners + plates	Approximately 5 per board, both layers	785	EA	Enhance perimeter / corners

Item	Basis / Calculation	Qty	Unit	Notes
Edge Metal and Termination				
Metal cap flashing / coping (10 ft)	Perimeter 237 LF x 1.05	25	pcs	Approximately 249 LF; pre-finished
Coping splice / cover plates	1 per joint	24	EA	Concealed splices
Coping corners (in / out)	Field count	5	EA	Prefab or fabricated
Termination bar (10 ft)	Perimeter 237 LF x 1.05	25	pcs	Top of base flashing
Drains and Penetrations				
Roof drain assemblies - "A"	Marker "A" count	2	EA	Clamp ring + strainer
Tapered sump kits	1 per drain	2	EA	Recommended - flat ISO
Curb / penetration flashing - "B"	Marker "B" - 39 in x 19 in	1	lot	TPO wrap + termination bar
Sundries and Accessories (Allowances)				
Bonding adhesive - base flashing	355 SF / 60 SF per gal	6	gal	Both attachment options
Pre-molded corners (in / out)	Detailing allowance	10	EA	Drains, curb, parapets
Water cut-off mastic	Detailing allowance	3	tubes	Drains, terminations
Cut-edge sealant (TPO)	Detailing allowance	3	bottles	Exposed cut edges

Optional - Cover Board (not included in totals): 1/2 in HD-ISO or gypsum cover board over the polyiso - approximately 83 boards (4 ft x 8 ft, field x 1.05). Recommended for improved puncture resistance, wind uplift, and warranty terms. Add if specified.

Membrane Attachment - Select One

Option A - Mechanically Attached	Lower Cost	Option B - Fully Adhered	Best Finish
Membrane fasteners + plates	252 EA	Field bonding adhesive	42 gal
Bonding adhesive (base flashing)	6 gal	Bonding adhesive (base flashing)	6 gal
Fasteners in the seams, with tighter patterns at perimeter and corner zones. Faster installation and lower material cost; wind performance is governed by the fastening pattern.		Membrane adhered across the field (approximately 60 SF per gal, both surfaces). Smoother finish and strong wind performance; higher adhesive cost and temperature-sensitive installation.	

Assumptions and Notes

Waste factors: 10% membrane, 7% insulation, 5% metal. Adjust for crew, roof complexity, and off-cuts.

Geometry: Field area modeled as a right trapezoid from the sketch. The 5 ft 6 in top-right clip removes less than 1% (approximately 15 SF) and is absorbed by waste; field-verify dimensions before ordering.

Wind and fastening: Fastener counts are baseline. Final pattern per ASCE 7 wind zone and FM / manufacturer requirements; enhance perimeter and corner zones.

Drainage: Two tapered sump kits are included since the polyiso is flat (no positive slope). Confirm tapered-ISO / slope scope separately if required.

Cover board: Excluded from totals; recommended for durability and warranty (approximately 83 boards if added).

Deck: Existing plywood assumed sound and fastenable; verify condition and fastener pull-out values.

Scope: Material planning take-off. Confirm all quantities against final approved drawings and the selected manufacturer's system requirements.